



Overview of Advanced Interpolation Methods

MAT 425

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OUTLINE

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- 1.1. Lagrange Interpolation
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4. MATLAB and Tractography

- 4.1. MATLAB
- 4.2. Tractography

Lagrange Interpolation

The first 1D interpolation method discussed was **Lagrange Interpolation**. It constructs a polynomial from a given set of data points to estimate values between them.

Given $n + 1$ points:

$$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$$

The interpolation polynomial is:

$$P(x) = \sum_{j=0}^n y_j \cdot L_j(x)$$

where the basis polynomial is:

$$L_j(x) = \prod_{\substack{0 \leq m \leq n \\ m \neq j}} \frac{x - x_m}{x_j - x_m}$$

Newton's Divided Difference

Newton's Divided Difference is a 1D interpolation method that uses a recursive table of divided differences to compute coefficients.

Given data points:

$$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$$

x_i	$f[x_i]$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$	$f[x_i, x_{i+1}, x_{i+2}, x_{i+3}]$
x_0	y_0			
x_1	y_1	$\frac{f[x_1] - f[x_0]}{x_1 - x_0}$		
x_2	y_2	$\frac{f[x_2] - f[x_1]}{x_2 - x_1}$	$\frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0}$	
x_3	y_3	$\frac{f[x_3] - f[x_2]}{x_3 - x_2}$	$\frac{f[x_3, x_2] - f[x_2, x_1]}{x_3 - x_1}$	$\frac{f[x_3, x_2, x_1] - f[x_2, x_1, x_0]}{x_3 - x_0}$

Cubic Spline Interpolation

Cubic spline interpolation fits a piecewise cubic polynomial between data points, ensuring smoothness by enforcing continuity of the first and second derivatives.

Conditions for cubic splines:

1. $S(x)$ is a cubic polynomial $S_j(x)$ on $[x_j, x_{j+1}]$ for $j = 0, \dots, n - 1$.
2. $S_j(x_j) = f(x_j)$, $S_j(x_{j+1}) = f(x_{j+1})$.
3. $S_j(x_{j+1}) = S_{j+1}(x_{j+1})$ function continuity.
4. $S'_j(x_{j+1}) = S'_{j+1}(x_{j+1})$ first derivative continuity.
5. $S''_j(x_{j+1}) = S''_{j+1}(x_{j+1})$ second derivative continuity.
6. Boundary conditions (choose one):
 - 6.1 $S''(x_0) = S''(x_n) = 0$ natural (free) BC.
 - 6.2 $S'(x_0)$ and $S'(x_n)$ specified clamped BC.

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Paper 2: Q-ball Imaging

Goal

The main goal of paper 2 is to propose a solution for Q-ball imaging reconstruction of the orientation distribution function (ODF).

- ODF: $G[S](\mathbf{u}) = \int_{|\mathbf{w}|=1} \delta(\mathbf{u}^T \mathbf{w}) S(\mathbf{w}) d\mathbf{w}$
- Estimate the function on a q-ball using spherical harmonics

Definition: A spherical harmonic of degree n denoted by $Y_n(\xi)$, is a harmonic homogeneous polynomial of degree n restricted to the unit 2-sphere.

$Y_n(\xi) = P_n|_{\mathbb{S}^2}$ which means

$$Y_n : \mathbb{S}^2 \rightarrow \mathbb{R}, \text{ such that } Y_n(\xi) = P_n(\xi) \forall \xi \in \mathbb{S}^2$$

Paper 2: Q-ball Imaging

Theory

- Approximated signal at each of the gradient directions:

$$S(\theta_i, \phi_i) = \sum_{j=1}^R c_j Y_j(\theta_i, \phi_i)$$

- They use $R = \frac{(l+1)(l+2)}{2}$
- This can be written as: $S = BC$
- Need to solve for coefficients:

$$c_j = \int_{\Omega} S(\theta, \phi) Y_j(\theta, \phi) d\Omega$$

Paper 2: Q-ball Imaging

Theory

- They add local regularization in order to use a high order estimation without over-modeling small perturbations with Laplace-Beltrami Operator
- Thus they define a measure, E , of the deviation from smoothness of a function f on the unit sphere as:

$$E(f) = \int_{\Omega} (\Delta_b f)^2 d\Omega$$

- Since the SH basis is orthonormal this can be written as:

$$E(f) = \int_{\Omega} \Delta_b \left(\sum_p c_p Y_p \right) \Delta_b \left(\sum_q c_q Y_q \right) d\Omega =$$

$$\sum_{j=1}^R c_j^2 l_j^2 (l_j + 1)^2 = C^T L C$$

Paper 2: Q-ball Imaging

Theory

- Matrix form: $M(C) = (S - BC)^T(S - BC) + \lambda C^T L C$
- $C = (B^T B + \lambda L)^{-1} B^T S$
- From this the SH coefficient of the signal on the q-ball for any (θ, ϕ) can be recovered as:

$$S(\theta_i, \phi_i) = \sum_{j=1}^R c_j Y_j(\theta_i, \phi_i)$$

Paper 2: Q-ball Imaging

Theory

- Replacing the signal by the SH series approximation:

$$G[S](\mathbf{u}) = \int_{|\mathbf{w}|=1} \delta(\mathbf{u}^T \mathbf{w}) S(\mathbf{w}) d\mathbf{w} = \sum_{j=1}^R c_j \int_{|\mathbf{w}|=1} \delta(\mathbf{u}^T \mathbf{w}) Y_j(\mathbf{w}) d\mathbf{w}.$$

- Funk-Hecke Theorem:** Let $f(t)$ be continuous on $[-1, 1]$ and H_l and SH of order l in C^3 . Then, given a unit vector $\mathbf{u}$

$$\int_{|\mathbf{w}|=1} f(\mathbf{u}^T \mathbf{w}) H_l(\mathbf{w}) d\mathbf{w} = \lambda(l) H_l(\mathbf{u}),$$

where

$$\lambda(l) = 2\pi \int_{-1}^1 P_l(t) f(t) dt$$

where P_l is the Legendre polynomial of degree l .

Paper 2: Q-ball Imaging

Theory

- Using Gaussian to approximate Dirac Delta Function:

$$\int_{|w|=1} \delta(\mathbf{u}^T \mathbf{w}) Y_j(\mathbf{w}) d\mathbf{w} = 2\pi P_l(0) Y_j(u),$$

where $P_l(0)$ is the Legendre Polynomial of degree l evaluated at 0.

- Using combinatorics techniques it can be obtained that when l is odd $P_l(0)$ is 0 and when l is even $P_l(0)$ is $(-1)^{\frac{l}{2}} \frac{1*3*5*\dots*(l-1)}{2*4*6*\dots*l}$. Now this can be applied to the ODF:

$$G[S](\mathbf{u}) = \sum_{j=1}^R 2\pi P_{l_j}(0) c_j Y_j(\mathbf{u})$$

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REFERENCES